Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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tikz,bm circuitikz

mycolorrgb1,0.2,0.3 brightgreenrgb0.4, 1.0, 0.0 britishracinggreenrgb0.0, 0.26, 0.15 cadmiumgreenrgb0.0, 0.42, 0.24 ceruleanbluergb0.16, 0.32, 0.75 darkelectricbluergb0.33, 0.41, 0.47 darkpowderbluergb0.0, 0.2, 0.6 darktangerinergb1.0, 0.66, 0.07 emeraldrgb0.31, 0.78, 0.47 palatinatepurplergb0.41, 0.16, 0.38 pastelvioletrgb0.8, 0.6, 0.79

ELEC 340 — Applied Electromagnetics and Photonics

Name: David Li
Student Number: V00818631
Due Date: January 25 2018, 11:30 AM
Assignment: Number 3

Problem 1 Given that the electric field in free space is:

$$E(R,\theta,t) = \hat{\theta}\frac{2}{R}\sin(\theta)\cos(6\pi\times10^9 t - 2\pi R)\ m/$$

where R and θ are the radial and polar variable in the spherical coordinate system. Find:

[label=()]The phasor representation of the given electric field vector.
$\tilde{\mathbf{E}}(R,\theta) = \hat{\theta}E_\theta = \hat{\theta}\frac{2}{R}\sin(\theta)e^{-j2\pi R}m/$ The phasor representation of the associated magnetic field vector.
$\nabla\times\tilde{\mathbf{E}} = j\omega\mu\tilde{\mathbf{H}}$
$\tilde{\mathbf{H}} = \frac{1}{j\omega\mu}\nabla\times\tilde{\mathbf{E}}\ \ \nabla\times\tilde{\mathbf{E}} = \frac{1}{R\sin\theta}\left(\frac{\partial}{\partial\theta}(E_\phi sin(\theta)) - \frac{\partial E_\theta}{\partial\phi}\right)\hat{\mathbf{R}} + \frac{1}{R}\left(\frac{1}{\sin(\theta)}\frac{\partial E_R}{\partial\phi} - \frac{\partial}{\partial R}(RE_\phi)\right)\hat{\theta}$
$+\ \frac{1}{R}\left(\frac{\partial}{\partial R}(RE_\theta) - \frac{\partial E_R}{\partial\theta}\right)\hat{\phi}$ Since $\tilde{\mathbf{E}}$ only has non-zero values in the $\hat{\phi}$ direction. $\nabla\times\tilde{\mathbf{E}} = \frac{1}{r\sin\theta}\left(-\frac{\partial E_\theta}{\partial\phi}\right)\hat{\mathbf{R}} - \frac{1}{R}\frac{\partial}{\partial R}(RE_\theta)\hat{\phi} = -\left(\frac{1}{R\sin\theta}\left(\frac{\partial E_\theta}{\partial\phi}\right)\hat{\mathbf{R}} + \frac{1}{R}\frac{\partial}{\partial R}(RE_\theta)\hat{\phi}\right)\ \ \tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla\times\tilde{\mathbf{E}} = \frac{-1}{j\omega\mu}\hat{\phi}\frac{0.002}{R}\sin\theta\frac{\partial}{\partial R}(e^{-j2\pi R})$
$= \hat{\phi}\frac{2\pi}{j6\pi\times10^9\times4\pi\times10^{-7}}\frac{0.002}{R}\sin(\theta)e^{-j2\pi R}$
$= \hat{\phi}\frac{5.30516477\times10^{-7}}{R}\sin(\theta)e^{-j2\pi R-\pi/2}\ (/)$
$= \hat{\phi}\ \frac{53}{R}sin(\theta)e^{-j2\pi R-\pi/2}\ (\mu/)$ The time-domain representation of the magnetic field you obtained in (b). $= \hat{\phi}\ \frac{53}{R}\sin(\theta)\cos(6\pi\times10^9 t - 2\pi R - \pi/2)\ (\mu/)$

Problem 2 The electric field intensity of a 5-MHz linearly polarized uniform plane wave traveling in free space in 10 V/m. The electric field is polarized in the +z direction at t = 0 and the wave in propagating in the -y direction. Find:

[label=()]The angular frequency and wave number, and intrinsic wave impedance.
Since the plane wave is traveling in free space, $\mu = \mu_0$ and $\epsilon = \epsilon_0$. $\omega = 2\pi f = 10\pi M\quad k = \omega\sqrt{\mu\epsilon} = 10\pi\times10^6\times3.335641\times10^{-9} = 0.0334\pi$ / The intrinsic impedance of a lossless medium is defined as: $\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \frac{\mu}{\epsilon}$ () Since free space is being used: $\eta = \eta_0 = 120\pi\Omega$
The field vectors phasor, i.e. $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$
$\tilde{\mathbf{E}} = -\hat{\mathbf{y}}E_0 = -\hat{\mathbf{y}}10e^{-j0.0334\pi z}/$
$\nabla\times\tilde{\mathbf{E}} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)\hat{\mathbf{z}}$
$\tilde{\mathbf{H}} = \frac{1}{j\omega\mu}\nabla\times\tilde{\mathbf{E}} = -\frac{1}{j\omega\mu}\frac{\partial E_y}{\partial z}\hat{\mathbf{x}} = \frac{10(0.0334\pi)e^{-j0.0334\pi z}\hat{\mathbf{x}}}{j\times2\pi\times10\times10^6\times4\pi\times10^{-7}}$
$= \hat{\mathbf{x}}\ \frac{0.08339}{2\pi}e^{-j0.0334\pi z-\pi/2}\ /$ E = -$\hat{\mathbf{y}}10\cos(2\pi10\times10^6 t - 0.0334\pi z)\ /$
$\mathbf{H} = \ \hat{\mathbf{x}}\ 0.0132719\cos(2\pi10\times10^6 t - 0.0334\pi z - \pi/2)\ /$

3. Draw a diagram to illustrate the field vectors and propagation direction.
[thick,darkpowderblue] plot[domain=0:4.5,samples=200] (,cos(deg(pi*)),0); [gray,thick,pastelviolet] plot[domain=0:4.5,samples=200] (,0,cos(d

$$\tilde{\mathbf{H}} = \frac{\nabla \times \tilde{\mathbf{E}}}{j\omega\mu} \qquad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

E_0 = electric field amplitude
H_0 = magnetic field amplitude
c = speed of light (3×10^8m/s)

μ_0 = magnetic permeability in a vacuum, $\mu_0 = 1.3 \times 10^{-6}\,\mathrm{N/A^2}$
ε_0 = electric permeability in a vacuum, $\varepsilon_0 = 8.9 \times 10^{-12}\,\mathrm{C^2/Nm^2}$

Problem 3 Suppose that a uniform plane wave is traveling in the +x direction in a lossless dielectric ($\mu_r = 1$) with the 100 V/m electric field in the +z direction. If the wavelength is 25 cm and the velocity of propagation is 2×10^8/. Find:

[label=()]The relative permittivity ϵ_r and impedance η of the medium. $\mathrm{u}_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}}$
$\epsilon_r = \frac{1}{\mu_0 \mu_r \epsilon_0 u_p^2} = \frac{(1)}{4\pi\times10^{-7}/\ 8.85\times10^{-12}/\ (2\times10^8\ /)^2} = 2.24795$
$\eta = \frac{\mu}{\epsilon} = \frac{\mu_r \mu_0}{\epsilon_r \epsilon_0} = \frac{4\pi\times10^{-7}\ /}{2.24795\times8.85\times10^{-12}\ /} = \frac{120\pi}{2.24795}\Omega = 53.38197\pi\ \Omega$ The angular frequency ω and the wave number k. Wave Number: $\mathrm{k} = 2\pi \frac{}{\lambda=\frac{2\pi}{0.25}=8\pi\ /\omega=\mu_p\times k=2\times10^8\ /\times8\pi\ /=16\pi\times10^8/}$
The time-domain expressions for the electric and magnetic field vectors. $\tilde{E} = \hat{\mathbf{x}}\tilde{E}_0 = \hat{\mathbf{x}}100e^{-jkz}\cos(\omega t - kz) = \hat{\mathbf{x}}100e^{-8\pi z}$ /
$\nabla \times \tilde{\mathbf{E}} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)\hat{\mathbf{z}}$
$\tilde{\mathbf{H}} = \frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}} = \frac{1}{j\omega\mu}\frac{\partial E_x}{\partial z}\hat{\mathbf{y}} = \frac{100(-8\pi)e^{-j8\pi z}\hat{\mathbf{y}}}{j\times16\pi\times10\times10^8\times4\pi\times10^{-7}} = -\hat{\mathbf{y}}\frac{1}{8\pi}e^{-j8\pi z - \pi/2}$

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3. $\mathrm{E} = \hat{x}E_0 = \hat{\mathbf{x}}100\cos(\omega t - kz) = \hat{\mathbf{x}}100\cos(16\pi \times 10^8 t - 8\pi z)$ /
$\mathbf{H} = -\hat{\mathbf{y}}\frac{1}{8\pi}\cos(16\pi \times 10^8 t - 8\pi z - \pi/2)$ /

4. Draw a diagram to illustrate the field vectors and propagation direction.

[x=(-10:1.75cm),y=(90:1.75cm),z=(210:1.75cm)] (-1,0,0) node[above] x – (5,0,0); (0,0,0) – (0,2,0) node[above] z; (0,0,0) – (0,0,2) node[left] $-y$; [->,ultra thick] (6,0,0) – node[above] c (5,0,0); [thick,red] plot[domain=0:4.5,samples=200] (,cos(deg(pi*)),0); [gray,thick] plot[domain=0:4.5,samples=200] (,0,cos(deg(pi*))); in 0.1,0.3,...,4.4 [->,help lines,brightgreen] (,0,0) – (,cos(deg(pi*)),0); [->,help lines] (,0,0) – (,0,cos(deg(pi*))); [above right] at (0,1,0) $E_0 = 100$; [below] at (0.55,0.15,1) $H_0 = 18\pi$;

$$\tilde{\mathbf{H}} = \frac{\nabla \times \tilde{\mathbf{E}}}{j\omega\mu} \qquad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

E_0 = electric field amplitude
H_0 = magnetic field amplitude
c = speed of light (3×10^8m/s)

μ_0 = magnetic permeability in a vacuum, $\mu_0 = 1.3 \times 10^{-6}\,\mathrm{N/A^2}$
ε_0 = electric permeability in a vacuum, $\varepsilon_0 = 8.9 \times 10^{-12}\,\mathrm{C^2/Nm^2}$